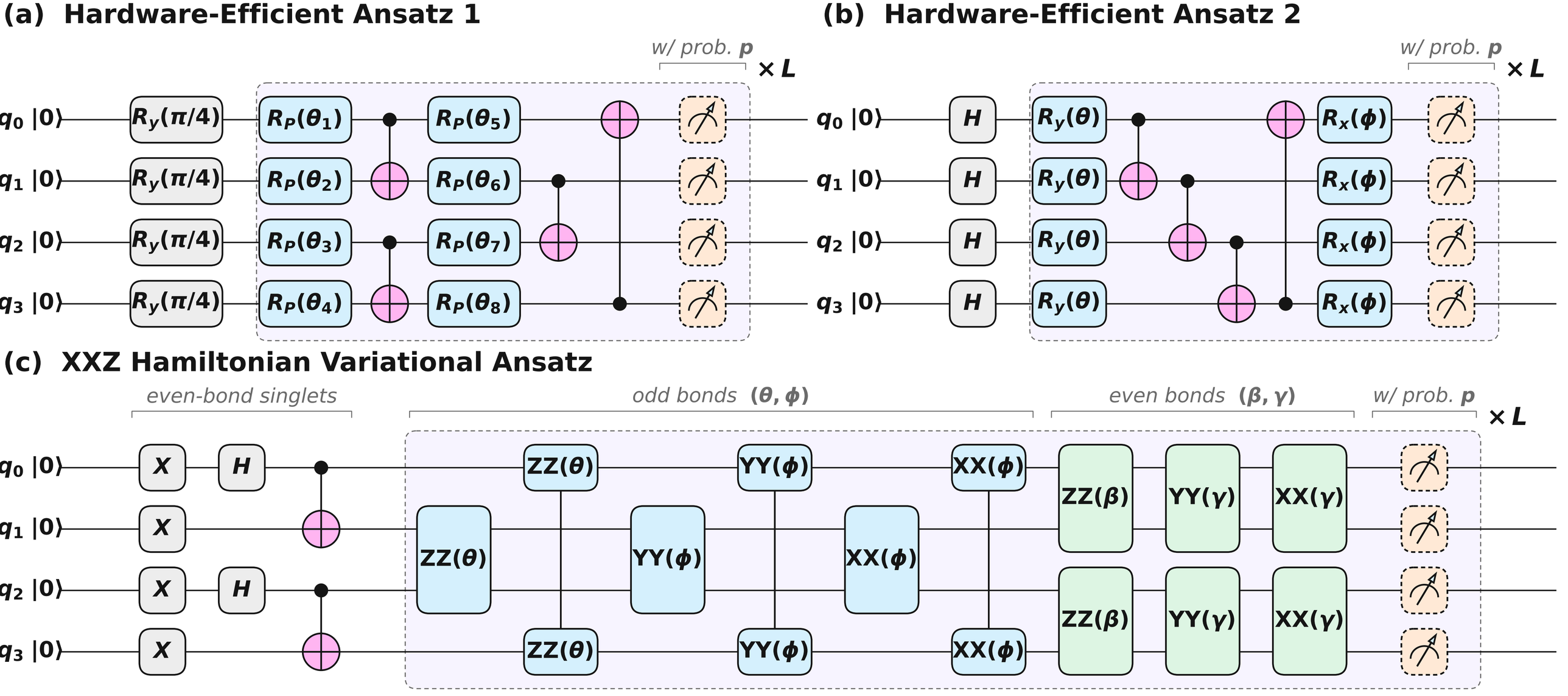
(a) Hardware-Efficient Ansatz 1
w/ prob. p
× L
q0 |0⟩
q1 |0⟩
q2 |0⟩
q3 |0⟩
Ry(π/4)
RP(θ1)
RP(θ2)
RP(θ3)
RP(θ4)
RP(θ5)
RP(θ6)
RP(θ7)
RP(θ8)
(b) Hardware-Efficient Ansatz 2
w/ prob. p
× L
H
Ry(θ)
Rx(ϕ)
(c) XXZ Hamiltonian Variational Ansatz
even-bond singlets
odd bonds (θ, ϕ)
even bonds (β, γ)
w/ prob. p
× L
X
H
ZZ(θ)
YY(ϕ)
XX(ϕ)
ZZ(β)
YY(γ)
XX(γ)